

A Spectrum- and Power-Efficient EHF Mobile Satellite System to be Integrated with Terrestrial Cellular Systems

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Abstract—An EHF satellite system for land-mobile applications to be integrated with a terrestrial cellular system is described in the paper. An approach to evaluate the carrier-to-cochannel interference occurring in a multispot satellite coverage adopting frequency reuse is introduced and results from the analysis are shown. Criteria for spectrum efficiency evaluation are also outlined along with traffic and link budget estimates. Possible options for payload implementation and mobile terminal design are presented.

I. INTRODUCTION

MOBILE radio market forecasts indicate a dramatic increase of user population in Europe during the 1990's, associated with the imminent introduction of the first pan-European digital cellular system, the GSM [1]. The GSM system has been conceived to essentially provide net 13 kb/s digital speech and 9.6 kb/s data communications at 900 MHz, with cell dimensions of a few kilometers and almost complete roaming capability over Western Europe—a distinct advantage over existing national systems. The scenario of mobile systems and services in Europe is expected to rapidly evolve thereafter and, in this evolution, the GSM is assigned the role of a first-generation system. A second pan-European system is already planned, the DCS1800, that will provide essentially similar services in the 1800 MHz frequency band with increased capacity. The Universal Mobile Telecommunications System (UMTS) is the following player, with the task to gradually supersede the pan-European systems family mentioned above.

The rapid evolution and deep penetration of the terrestrial mobile and personal services, however, are likely to be characterized by some limitations. In particular, the complete set of ISDN services, which is not expected to be provided by the GSM system in the near future, will presumably be introduced with additional delay by the

UMTS in rural and suburban areas. In fact, the basic ISDN access 2B + D (two 64 kb/s channels and one 16 kb/s channel) cannot be transparently connected to a GSM channel but will be provided by the UMTS, preeminently in urban environments at first, where small cells are economic so that the high channel data rates that are necessary to combat channel impairments can be achieved [2]. In suburban and rural environments, the satellite stands as a cost-effective candidate, provided that its functions are integrated with the more developed terrestrial networks.

At present, some autonomous satellite systems devoted to terrestrial services are being deployed in Europe: the European Space Agency (ESA) is developing two land-mobile *L* band experimental payloads—the EMS to be embarked on ITALSAT II (1994) and the LLM to be embarked on ARTEMIS (1995). Both systems will provide voice communications with European coverage [3].

Full system integration with the GSM system (i.e., the capability to design a satellite system according to the GSM "air-interface" specifications) is extremely difficult to achieve [4]. However, if network integration is the objective, a satellite system can be designed with different transmission characteristics but share at least part of the fixed facilities of an existing terrestrial cellular system. Under this assumption, the two systems are distinct but share network functions, thus providing a transparent ISDN service to the mobile user equipped with a suitable "dual-mode" terminal that automatically routes signals via satellite or terrestrial facilities. In this case, the satellite spot coverage areas can also be functionally considered cells even if they are of a larger size, and the satellite lattice coverage overlaps the terrestrial cellular layout providing it with extended coverage.

In earlier papers [5], [6], the authors proposed the combined use of the EHF (more specifically, 40/50 GHz) band and Highly Elliptical Orbits (HEO) to implement a new satellite system integrated with the GSM system. Since the *L* band is very congested, it is mandatory to exploit higher frequencies and, in perspective, the EHF seems to be the most suitable candidate. In fact, it is well known that difficult propagation conditions can impose extremely high power margins at 40/50 GHz (even several tens of dB) when dealing with very high availability

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values (on the order of 99.9% or higher). Under these conditions, the use of the EHF band is certainly discouraged. On the other hand, it has been demonstrated [7] that sufficient link margins can be achieved, even at 50 GHz, if the link availability requirement is not too stringent, e.g., about 99%, which is acceptable in the case of mobile systems.

Highly elliptical orbits with the apogee on the zenith of Central Europe minimize the probability of link obstruction, which is another important contribution to the overall system unavailability. In the case of mobile communications, several HEO's have been proposed (e.g., see [8]) with an inclination of 63.4° over the equatorial plane.

The joint adoption of satellites located on HEO's and of the 40/50 GHz band presents several implications at system level that have been evaluated [5]. Among them, the main aspects are: handover between satellites and cells, on-board antennas zooming effect due to satellite motion, the Doppler effect, and problems related to the high directivity of the mobile terminal antenna. A preliminary link design of the EHF satellite system indicated that a bit error probability of 10^{-6} and a system availability of about 99% are achievable out of metropolitan areas. European coverage is achieved via a multibeam system.

In this paper, we illustrate the selected concept of an integrated satellite and terrestrial cellular system (Section II). Then, we dwell upon the main criteria in system design (Section III). In particular, the satellite system is not simply a noise-limited system, but is intended to operate in the presence of relatively high levels of cochannel interference. Since a direct system performance optimization in the presence of both effects is difficult, we prefer to concentrate on the two limiting conditions of prevailing noise and prevailing interference, giving the system a reasonable margin for combined effects. With respect to the interference-limited system condition, we introduce an approach to evaluate the carrier-to-cochannel interference ratio and provide results of the analysis. Then, we report some typical values for a system design (Section IV). Finally, with a look to feasibility, we indicate some possible alternatives for payload implementation and some design criteria for the mobile terminal phased-array antenna (Section V).

II. INTEGRATION OF SATELLITE AND TERRESTRIAL CELLULAR SYSTEMS

As is well known, the essential features of a cellular system are frequency reuse by spatial separation and capability of cell splitting [9]. By applying these two concepts, a mobile system progressively grows in subscriber capacity without any need for additional spectrum allocation, thus dramatically improving during its lifetime even an initially poor spectrum efficiency. In principle, the cell splitting process, with associated frequency reuse, can be applied an indefinite number of times so that the spectrum efficiency growth can be unbounded. In practi-

cal vehicular cellular systems, however, a minimum cell radius must be assumed to limit the probability of call handover.

Frequency reuse is customary in the satellite systems practice. However, in present geostationary *L*-band mobile systems it is hardly implemented. To achieve a moderate degree of frequency reuse at the European level, very large on-board antennas are required. For example, according to the present design, the ARTEMIS LLM payload will only provide three spots over Europe with about 3 m diameter antenna reusing the frequency in the two external spots. A much higher value for the frequency reuse factor can be achieved at 40 GHz. In this case, Western Europe may be covered with about 40 spots with an on-board antenna diameter of only 0.5 m.

Cell splitting is not so easily implemented as in terrestrial systems, where fixed facilities can be installed and relocated according to traffic variations. Starting from an initial system configuration with small-sized satellites, in principle cell splitting is achievable with additional satellites exactly as in terrestrial systems, however with some earlier planning. In practice, for a cost-effective system, we can transfer to the satellite world the cell splitting concept with some limitations. First, taking into account an uneven traffic distribution at the Continental level the cell arrangement can be planned to be differently sized. Second, when a set of adjacent cells (generally not a single one) served by a first-generation satellite approaches saturation, a second-generation satellite equipped with more directive antennas capable of splitting (e.g., in three or four parts) each original cell can be put into operation, which will reuse channel sets used sufficiently far apart. In the example of a European first-generation EHF system with 40 cells and limited frequency reuse, by gradually replacing original satellites with new ones accommodating 1 m antennas and rearranging the frequency assignment, coverage is gradually accomplished with more and more cells (about 160 cells, in the limit) thus achieving a remarkable increase in spectrum efficiency.

In a cellular system, each cell is supposed to be served by a base station (BS). In an integrated terrestrial/satellite system, it is possible to achieve a dramatic reduction in the overall system complexity if the terrestrial system can consider every satellite cell as an additional macrocell superimposed to its cellular layout. Inside the macrocell, the mobile terminal is connected via satellite to an earth station which acts as an access port to the terrestrial network. Fig. 1 shows the integrated satellite/terrestrial cellular coverage. If we can assume that the existing GSM fixed network facilities (trunks and switching centers) can carry the additional satellite traffic with marginal cost, on-board routing is not necessary in principle. To this purpose, we note that the GSM fixed facilities are designed to carry true 64 kb/s traffic bundles in spite of the lower net user data rate adopted in the GSM radio link. As a consequence, the role of the satellite system is limited to providing the GSM with an additional set of special base stations (SBS) so that a minimum complexity system configuration can be achieved. Each SBS which serves a satel-

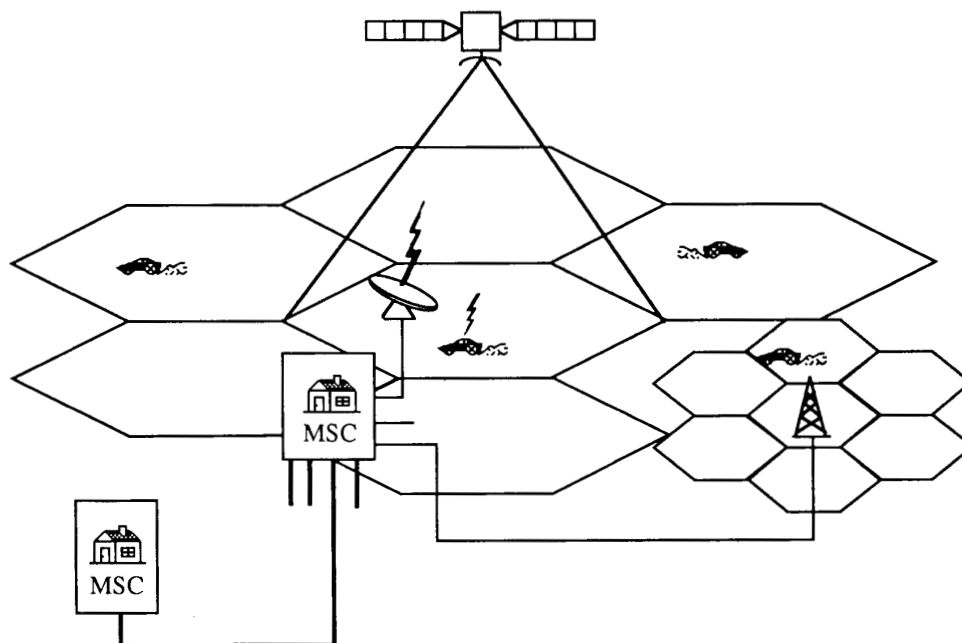


Fig. 1. Integration of a cellular satellite system with the GSM system.

lite cell includes: a) one or more fixed earth stations (FES), located within the cell and connected to a GSM mobile switching center (MSC); b) the transponder resource needed to cover the cell itself.

III. CHANNEL CHARACTERISTICS

A. Satellite Channel

To provide service to mobiles located in high European latitudes, highly elliptical orbits are most suitable, having an inclination of 63.4° over the equatorial plane. By a proper selection of orbital parameters, the satellite track projected on ground is independent of time. Fig. 2 shows a three-satellite system located on three HEO's separated by 120° . Fig. 2(a) and (b) show the satellite's constellation orbits as seen by a fixed and Earth observer, respectively. Fig. 2(c) shows the projection on ground of the satellites tracks. Among various possible orbital choices, the Loopus orbits family has the remarkable advantage of a single node for each of the stationary loops, as shown in Fig. 2(c). In the case of two loops, satellites can be phased so that when a satellite is at its apogee within a loop, the other two satellites are located at the node of the other loop, one entering and the other exiting the loop [8], [10]. An important figure of merit is the elevation angle of the satellite within the coverage area. With the adoption of highly elliptical orbits, Europe can be served with elevation angles higher than 55° – 60° and higher than 65° – 70° in the case of Central and Northern Europe, thus achieving a marked advantage over the geostationary orbit.

According to Radio Regulations, an EHF mobile satellite system can be allocated in the 40/50 GHz band.

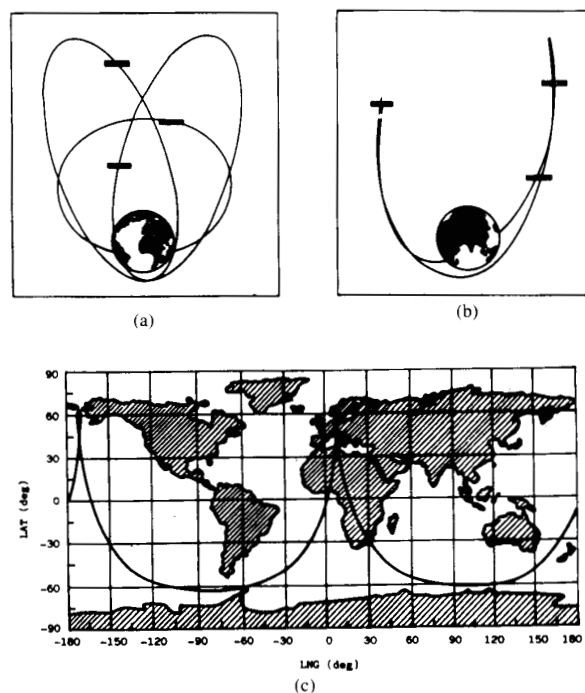


Fig. 2. Satellite constellation orbits as seen from: a) a fixed observer; b) an Earth observer; c) projected on ground.

The 40/50 GHz mobile radio channel mainly suffers from rain and snow effects, clouds and fog, water vapor, molecular absorption, and tropospheric turbulences. As was noticed from data extrapolated from the Sirio and OTS experiments [5], [7], for 1% unavailability rain is the ma-

jor effect but it introduces limited attenuation values, generally comparable with those due to all other contributions.

The use of highly directive mobile terminal antennas strongly reduces the multipath effects. Therefore, in the majority of sites the channel can be considered either additive white Gaussian noise or fully obstructed by obstacles. The satellite radio visibility can be considered a binary event and, in the absence of cochannel interference, the outage probability can be derived through the evaluation of the probability of propagation anomalies, P_{blo} , and the attenuation probability due to atmospheric effects, P_{atm} :

$$P\{\text{outage/no interference}\} = P_{atm} + P_{blo} - P_{atm}P_{blo}. \quad (1)$$

P_{atm} is a function of the propagation margin while P_{blo} is almost independent of the satellite system characteristics (e.g., EIRP). The unavailability seems to be mostly caused by atmospheric effects, if the system is assumed to operate essentially in suburban and rural areas where the blocking factor is generally small [5].

The target of 1% for the overall unavailability of the 40/50 GHz system seems reasonable in many European sites. More thorough evaluations, however, need for further experimental data in terms of atmospheric and environmental effects at 40/50 GHz in the case of highly elliptical orbits.

B. Interference Analysis

Even if the selected 40/50 GHz frequency band is currently unused, its efficient use is at least dictated by technological and economic reasons connected to the feasibility of a single wideband mobile terminal capable to operate in the whole service area. As a consequence, frequency reuse should be adopted from the beginning of system design, which then necessitates an interference analysis. In the long term, frequencies could be more intensively reused and on-board antenna gains and output power levels will presumably increase, thus increasing the relative weight of cochannel interference. In the following paragraphs, some models are introduced for the evaluation of the interference effects in a multispot satellite system scenario which adopts frequency reuse.

In a classic cellular system, the carrier-to-interference ratio C/I is a function of the cluster size m (e.g., $m = 3$, see Fig. 3). The N available channels are divided into m subsets and each subset is assigned to a different cell of the cluster. The cluster size determines the ratio between the so-called "cell radius" r and the minimum distance d between the centers of cochannel cells, i.e.,

$$\frac{d}{r} = \sqrt{3m}. \quad (2)$$

Therefore, $C/I = f(d/r)$ and the behavior of this function depends on the channel characteristics.

In a satellite system, C/I still depends on d/r by means of the on-board antenna radiation pattern, which acts as a

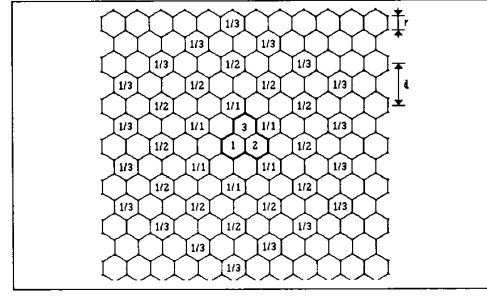


Fig. 3. Cellular coverage with cluster size $m = 3$ and the first three interfering tiers.

"spatial filter" to allow the reuse of channels over the service area. According to the cellular layout, the interfering cells can be divided into different tiers as shown in Fig. 3 [11]. The number of cells in the k th tier is equal to $6k$. For the sake of simplicity, let the distances from the center of the wanted cell to the centers of the interfering cells of the k th tier be approximately equal to kd . Let us assume return links from the mobiles to the fixed earth stations; let us also assume that the mobiles transmit equal power levels and that all interferers are simultaneously active. First, we introduce the general model for the C/I experienced in the return link from the mobile to the fixed earth terminal. Second, we provide some results in the case of transparent satellites with a deterministic set of system parameters. Finally, we turn to a statistical approach providing additional results.

1) *Carrier-to-Interference Ratio Modeling*: For transparent transponders in the worst case, the carrier-to-interference ratio C/I is

$$\frac{C}{I} = \frac{1}{\left(\frac{C}{I}\right)_1^{-1} + \left(\frac{C}{I}\right)_2^{-1}} \quad (3)$$

with

$$\left(\frac{C}{I}\right)_1 = \frac{\beta_w G_0(\theta_w)}{\sum_{k=1}^{N_i} \sum_{i=1}^{6k} \beta_{ik} G_0(\theta_{ik})} \quad (4)$$

and

$$\left(\frac{C}{I}\right)_2 = \frac{\beta_w G_0(0) G_0(\theta_w)}{\sum_{k=1}^{N_i} \sum_{i=1}^{6k} \beta_{ik} G_{ik}(\Theta_{ik}) G_{ik}(\theta_k)} \quad (5)$$

where $G_0(\theta)$ is the antenna gain of the spot pointed toward the cell where the wanted (i.e., interfered) mobile is located, $G_{ik}(\theta)$ is the antenna gain of the spot pointed toward the i th interfering cell in the k th tier, θ_w is the off-axis angle under which the wanted mobile is illuminated by $G_0(\theta)$, θ_{ik} is the interferer off-axis angle relative to the boresight of $G_0(\theta)$, θ_k is the off-axis angle of the center of a cell in the k th tier relative to the boresight of $G_0(\theta)$, Θ_{ik} is the interferer off-axis angle relative to the boresight of

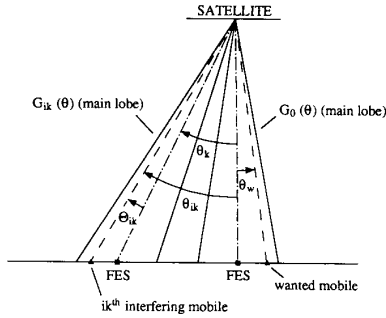


Fig. 4. Angular parameters utilized in the model.

$G_{ik}(\theta)$, N_t is the number of considered tiers and, finally, β_w^{-1} and β_{ik}^{-1} are the uplink excess attenuations for the wanted and interfering signals, respectively (see Fig. 4). In (4) and (5), each spot is pointed toward the center of its respective cell where the fixed earth station (FES) is supposed to be located.

In the case of regenerative satellites, the interference separately acts on the uplink and on the downlink. For the uplink, we find that the carrier-to-interference ratio $(C/I)_U$ is again expressed by (4). For the downlink, we have

$$\left(\frac{C}{I}\right)_D = \frac{G_0(0)}{\sum_{k=1}^{N_t} \sum_{i=1}^{6k} G_{ik}(\theta_k)}. \quad (6)$$

Under reasonable assumptions, the carrier-to-interference ratio $(C/I)_D$ experienced in the downlink of a regenerative satellite system is about 3 dB higher than $(C/I)_U$. Therefore, the uplink is the worst-case condition.

We also introduce the normalized far-field on-board radiation pattern $F(\theta)$, which is related to $G_0(\theta)$ by

$$G_0(\theta) = G_M F^2(\theta) \quad (7)$$

where G_M is the maximum gain. Similarly, we can introduce the interfering beam radiation pattern. If actual antenna patterns were adopted with their lobed behavior, the results would be unduly sensitive to mobile terminals position. Therefore, we prefer to rely upon reasonable antenna masks. Unfortunately, standard masks are not available for multispot cellular satellite coverage. To achieve realistic results, we derive the required masks based on generic tapered-aperture antennas, i.e., antennas having an aperture field intensity

$$E_a(\rho) = E_0 \left[(1 - T) + T \left(1 - \frac{4\rho^2}{d_a^2} \right)^p \right] \quad (8)$$

where d_a is the aperture diameter, ρ is the polar axis in the aperture plane, $p = 0, 1, 2, \dots$, and T is the aperture edge taper. In the case $T = 0$, the antenna aperture field is uniform. For the same value of p , larger values of $T_{dB} = -10 \log(1 - T)^2$ result in lower sidelobe levels, larger beamwidths, and lower illumination efficiencies. The

beam radiation pattern has the following expression [12]

$$F(u) = \frac{(p+1)(1-T)}{(p+1)(1-T) + T} \cdot \left(2 \frac{J_1(u)}{u} + 2^{p+1} p! \frac{T}{1-T} \frac{J_{p+1}(u)}{u^{p+1}} \right) \quad (9)$$

where

$$u = \frac{\pi d_a}{\lambda} \sin \theta$$

and $J_p(u)$ is the Bessel function of the first kind and order p . Antenna masks were constructed by means of broken lines which envelope the sidelobes of (9).

2) *Deterministic Analysis*: Let us now concentrate on transparent transponders and assume the following deterministic model:

- interfered mobile located on the border of the served cell (i.e., 3 dB coverage at satellite handover [5]);
- interferers located on the center of their cells;
- all links under normal propagation conditions, i.e., $\beta_{ik} = \beta_w$ (clear sky and no multipath fading).

Therefore, $\theta_{ik} = 0$ and $\theta_{ik} = \theta_k$ in (4) and (5). Under the additional assumption of equal spot antennas, i.e., $G_{ik}(\theta) = G_0(\theta)$, $(C/I)_1$ results equal to $(C/I)_2$. Then, we have

$$\frac{C}{I} = \frac{\gamma}{12} \frac{G_0(0)}{\sum_{k=1}^{N_t} k G_0(\theta_k)} = \frac{\gamma}{12} \left[\sum_{k=1}^{N_t} k F^2(u_k) \right]^{-1} \quad (10)$$

with $G_0(\theta_w) = \gamma G_0(0)$ (e.g., $\gamma = 0.5$ when $\theta_w = \theta_{3dB}/2$). By noting that $\sin \theta_k \cong \theta_k \cong kd/h$, where h is the satellite height, and that the beam half-power beamwidth θ_{3dB} (rad) $\cong 0.39 \pi \lambda / d_a$, where d_a is the on-board antenna diameter, we easily find

$$u_k \cong ku_0 \frac{d}{r} \quad (11)$$

where u_0 is a constant that equals $u_{3dB} \cong 1.29$ when $\theta_k = \theta_{3dB}/2$. This confirms that $C/I = f(d/r)$ for the satellite cellular system.

A similar deterministic model for the forward link shows that $(C/I)_1$ is 3 dB higher than (4) and $(C/I)_2$ is generally more than 3 dB higher than (5), if the sidelobe peaks of $F(u)$ monotonically decrease with u . In conclusion, we are given confidence on the correctness of our choice of the return link as the worst-case.

Some curves of C/I as a function of d/r [see (10)] are shown in Fig. 5 for four different antenna masks. The curves were obtained considering only the first tier of interferers and the satellite located at the apogee. For $d/r \geq 4$, the interferers are located out of the main lobe of $F(u)$. The dependence of C/I on N_t , shown in Fig. 6, refers to a particular choice of on-board antenna parameters. The influence of satellite height on C/I is not immediately evident since both the wanted mobile and the

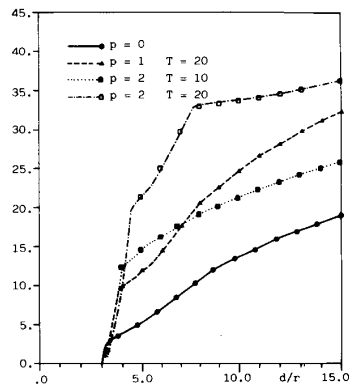


Fig. 5. C/I for fixed locations of the wanted mobile and of the interferers for different antenna masks ($N_t = 1$, satellite at apogee).

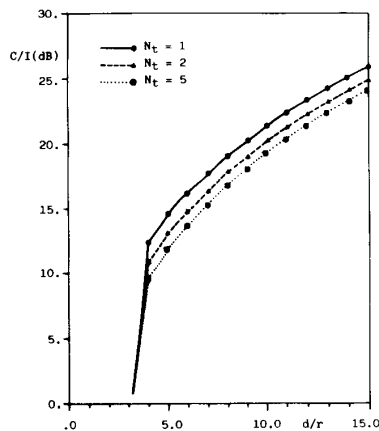


Fig. 6. C/I for fixed locations of the wanted mobile and of the interferers as a function of the number of interfering tiers ($p = 2$ and $T = 10$ dB, satellite at apogee).

interferers off-axis angles decrease when the satellite height increases, thus producing contrasting effects. However, it was verified that generally the increase of the interference level is prevailing; therefore, the satellite located at apogee is the worst-case condition.

3) *Statistical Analysis:* In a statistical approach (3), (4) and (5) can be retained, however considering θ_w , θ_{ik} , Θ_{ik} , β_w and β_{ik} ($k = 1, 2, \dots, N_i$; $i = 1, 2, \dots, 6k$) as random variables so that the probability distribution function of C/I , $P_{C/I}(x)$, can be evaluated accordingly. Let us first assume both β_w and $\{\beta_{ik}\}$ equal to unity (normal propagation) and a uniformly distributed location of both the wanted and interfering mobiles; $P_{C/I}(x)$ can be estimated by means of a computer simulation analysis. Fig. 7 shows the result of a simulation as a function of d/r , for three different values of the conditional outage probability

$$P\{\text{outage/no noise}\} = P_{C/I}(x_{Th}) \quad (12)$$

where x_{Th} is the minimum acceptable value for C/I to achieve a given transmission quality. In Fig. 7, antenna masks are assumed in accordance with (9) with $p = 2$ and

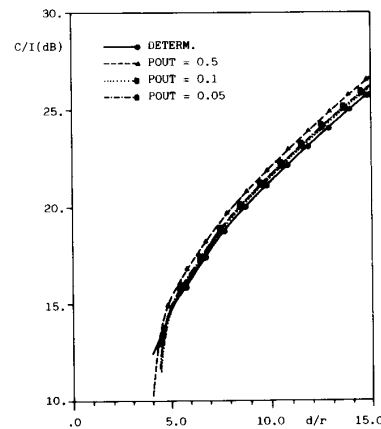


Fig. 7. C/I for random uniformly distributed positions of the wanted mobile and of the interferers for different values of the outage probability ($N_t = 1$, $p = 2$, and $T = 10$ dB, satellite at apogee).

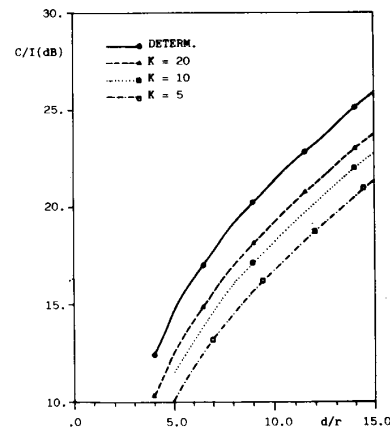


Fig. 8. C/I for a Rician channel for different values of the Rice factor K , assuming both the wanted mobile and the interferers are stationary ($N_t = 1$, $P_{out} = 0.1$, $p = 2$, and $T = 10$ dB, satellite at apogee).

$T = 10$ dB, $N_t = 1$, and the satellite located at apogee. The C/I computed with the deterministic model is also shown for comparison.

Now let us consider the excess uplink attenuations β_w^{-1} and $\{\beta_{ik}^{-1}\}$. They can originate from propagation anomalies such as blockage and strong multipath or from atmospheric effects. The latter effect is put into account in the system design by means of a fixed power margin. Signal blockage is considered negligible (see Section III-A). Specular multipath effects are also considered negligible due to the highly directive terminal antenna. On the contrary, the effect of diffuse multipath can be taken into account considering a statistical model, where a Rician distribution is assumed for the square roots of β_w and $\{\beta_{ik}\}$ while θ_w , θ_{ik} , Θ_{ik} are computed according to the deterministic model. The random variable C results in a noncentral χ^2 distribution with two degrees of freedom while, if only the first tier of interferers is considered, I is the sum of six independent r.v.'s, each representing the contribution of an interferer to the total interfering

power. Since all of these r.v.'s are noncentral χ^2 with two degrees of freedom and the same moments, I has a noncentral χ^2 distribution with twelve degrees of freedom [13]. Based on the probability density function of both C and I , we can numerically evaluate the probability distribution function of the C/I ratio [14]. It is worth noting that to completely describe a noncentral χ^2 r.v., it is sufficient to specify the "Rice factor" K and the mean value. While the latter can easily be determined for all considered variables, for the first only some assumptions can be made because of the lack of experimental results. In the literature, $K = 10$ is considered typical at L band. Fig. 8 shows some curves computed for three different values of K .

IV. SYSTEM DESIGN ELEMENTS

A. Spectrum Efficiency of Different Modulation Schemes

Spectrum efficiency can be defined as [15], [16]

$$\eta = \frac{MR}{W} \quad (13)$$

where R (bit/second) is the information data rate of a single channel, M is the user population, and W is the bandwidth allocated to the service. So defined, the spectrum efficiency η takes implicitly into account the three efficiency components: time efficiency η_t (due to trunking), space efficiency η_s (due to the channel reuse), and frequency efficiency η_f (due to modulation). The frequency efficiency can be expressed as

$$\eta_f = \frac{R}{W_0} \quad (14)$$

where W_0 is the ratio W/N between the available bandwidth and the maximum number of channels which can be allocated for a given multiple access and modulation scheme capable of supporting R . The value of η_f is easily fixed if the system is supposed to operate according to the Erlang B model (in the computations, a blockage probability of 0.01 and a user traffic of 0.02 Erlang are considered).

It is possible to determine the relationship between the required C/I , η_f , and η . Fig. 9 shows C/I versus η_f with the spectrum efficiency η as a parameter, assuming the deterministic model. The values are normalized to the value of the curve $\eta = \text{constant}$ which passes through a reference point ($C/I = 0$ dB, $\eta_f = 1$). Every modulation and coding system can be represented in the plane by the point (C/I , η_f) identified both by the minimum acceptable value of the C/I for a given bit error probability and by its frequency efficiency η_f . The modulation and coding system efficiency is that of the line $\eta = \text{constant}$ passing through the point.

Some well-known systems are quoted in Fig. 9, assuming that interference only is present and that it produces the same effect of a Gaussian noise on the bit error probability (10^{-6} in the figure). All the points have been derived referring to the Nyquist band. This figure allows

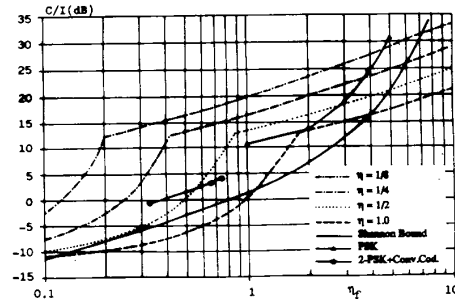


Fig. 9. Spectrum efficiency of different modulation and coding systems.

comparisons among different systems. To this purpose, we need to select one of the two representative points, to draw the curve $\eta = \text{constant}$ which passes through it, and observe the relative position of the other. In conclusion, the most efficient modulation/demodulation scheme for a selected scenario can be determined.

B. Traffic Estimates

Let us now present some numerical examples. Assuming Western Europe as the service area, the cellular layout is completely defined by the following parameters: the cell dimension that can be scaled by varying the on-board antenna diameter d_a and the cluster size m . The total number of cells over the service area N_{cell} immediately follows and $N_{\text{cluster}} = N_{\text{cell}}/m$ gives the space efficiency. The frequency band assigned to the service (e.g., 100 MHz) is statically partitioned between the cells. Inside a cell, channels are dynamically assigned on demand (e.g., SCPC/DAMA system). We assume to allocate 500 channels in the total frequency band; the number of channels per cell, n_{cell} , is a function of m . In calculating the total carried traffic T_{car} , we assume the same values for blockage probability and user traffic as used in Section IV-A.

Table I summarizes the main results of the traffic analysis for three different values of d_a and two values of m . In addition to the total carried traffic, Table I provides an indication of the total number of users M serviced in the peak hour. Increasing d_a increases the total number of cells and, hence, the carried traffic. However, larger antennas increase satellite complexity and a larger number of clusters increases the number of transponders necessary to achieve the coverage. Increasing m decreases the carried traffic since we lower the space efficiency. On the other hand, larger clusters improve C/I , thus alleviating the link budgets.

C. Link Budgets

Table II shows sample link budgets evaluated under the following main assumptions:

- satellite located at the apogee of a Loopup HEO;
- propagation unavailability $P_{\text{atm}} = 1\%$ ($P_{\text{bio}} = 0$);
- bit error probability $= 10^{-6}$;
- cluster size $m = 7$;
- information data rate $= 64$ kb/s;

TABLE I
TRAFFIC ESTIMATES AND USER POPULATION

$m = 7$				
$d_a(m)$	N_{cell}	n_{cell}	$T_{\text{car}}(E_{\text{rl}})$	M
0.5	40	72	2296	115000
1.0	157	72	9012	451000
2.0	650	72	37310	1865000
$m = 12$				
$d_a(m)$	N_{cell}	n_{cell}	$T_{\text{car}}(E_{\text{rl}})$	M
0.5	40	42	1220	61000
1.0	157	42	4788	239000
2.0	650	42	19825	991000

TABLE II
SAMPLE LINK BUDGETS

	Forward Link (Satellite to Mobile) $f = 40 \text{ GHz}$	Return Link (Mobile to Satellite) $f = 50 \text{ GHz}$
EIRP (dB W)	44.2 ⁽¹⁾	35.0
Free Space Attenuation (dB)	216.4	218.4
Excess Attenuation (dB)	5.2	9.4
G/T (dB/K)	5.0	20.4
Boltzmann Constant (dB W/HzK)	-228.6	-228.6
C/N_0 (dB Hz)	56.2	56.2
C/I (dB) ⁽²⁾	16.5	13.5
$(C/N_0)_{\text{TOT}}$ (dB Hz)	55.9	55.6
E_b/N_0 (dB)	7.9	7.6
$(E_b/N_0)_{\text{REQ}}$ (dB) ⁽³⁾	4.8	4.8
Margin (dB)	3.1	2.8

Notes:

(1) $d_a = 1 \text{ m}$, antenna efficiency = 0.65, gain at HPBW, $P_{\text{OUT}} = -4 \text{ dB W}$

(2) Antenna tapering: $T = 10 \text{ dB}$, $p = 2$

(3) BPSK, convolutional encoding ($R = 1/2$, $k = 7$)

- frequency: 40 GHz (satellite-to-mobile); 50 GHz (mobile-to-satellite);
- C/N on the fixed links much greater than C/N on the mobile links;
- C/I evaluated in accordance with the deterministic model;
- system margin = 3 dB;
- on-board antenna diameter $d_a = 1 \text{ m}$.

The same link budgets are still valid under the above-mentioned assumptions if we consider an on-board antenna diameter of 0.5 m or 2 m with a mobile terminal G/T of 11 dB/K or -1 dB/K , respectively. These two conditions can be considered the technological limiting cases for the mobile terminal and for the on-board antenna subsystem, respectively.

V. TECHNOLOGICAL ISSUES

A. On-Board Payload

The payload configuration to implement the satellite cellular concept is not unique. To be specific, let us assume that a payload is designed to serve a cellular cluster

so that it is associated to m antenna beams per transmission direction in the mobile link. In principle, both transparent and regenerative architectures are adequate, provided that they make an efficient use of the power generated on-board. The regenerative option needs for multicarrier demodulators that penalize it in terms of power consumption, weight, physical dimensions, and complexity. These disadvantages are hardly counterbalanced by the advantages in terms of better C/I and C/N performance. The tradeoff could be offset in a satellite system in which on-board routing functions must be accommodated.

Two payload architectures were already described in the literature that are applicable to a 40/50 GHz cellular system: the first is based on the use of the multiport amplifier (MPA) [17], while the second adopts the imaging antenna (IA) concept [18]. Both solutions must be implemented by means of active antennas to avoid serious impacts on EIRP due to beam-forming network losses. The feasibility of active antennas was already demonstrated at K_a band [19].

Apart from the efficient use of power, both the MPA and IA offer additional advantages in terms of intrinsic soft-fail capability and a certain degree of flexible routing. An evolution of the MPA concept, called a multiport transponder (MPT) [6], consists of m transponders in a parallel configuration located between the input and output branching networks, and provides some additional advantages in terms of flexible routing within a cell or among adjacent cells.

A preliminary comparison between the MPA and IA solutions shows that the IA needs a dual-reflector antenna configuration, with penalties in terms of weight and dimensions that are counterbalanced by a better antenna efficiency and the capability to compensate the zooming effect. On the other hand, the MPA could be associated to a single reflector antenna with a relatively simple cluster of circular or pyramidal feed-horns, one feed per beam. The main disadvantage of the MPA is its output passive network that lowers the overall antenna efficiency. The above indicates both configurations should have their range of applicability depending on the cluster size.

B. Mobile Terminal

The development of a high-performance small-size low-cost user terminal is a very challenging technological task [20], essentially due to the need for tracking the satellite in a 60° cone [5], which is imposed by the use of HEO's, and to the use of millimeter-wave circuitry.

The most suitable configuration for the antenna seems to be an electronically steerable active phased array. Planar radiating elements should be used to allow low-cost large-scale production. Dual frequency operation can be avoided by separating the transmitting and receiving antennas at the expense of a moderate increase in accommodation complexity but still holding very acceptable terminal dimensions. A possible configuration for the mobile

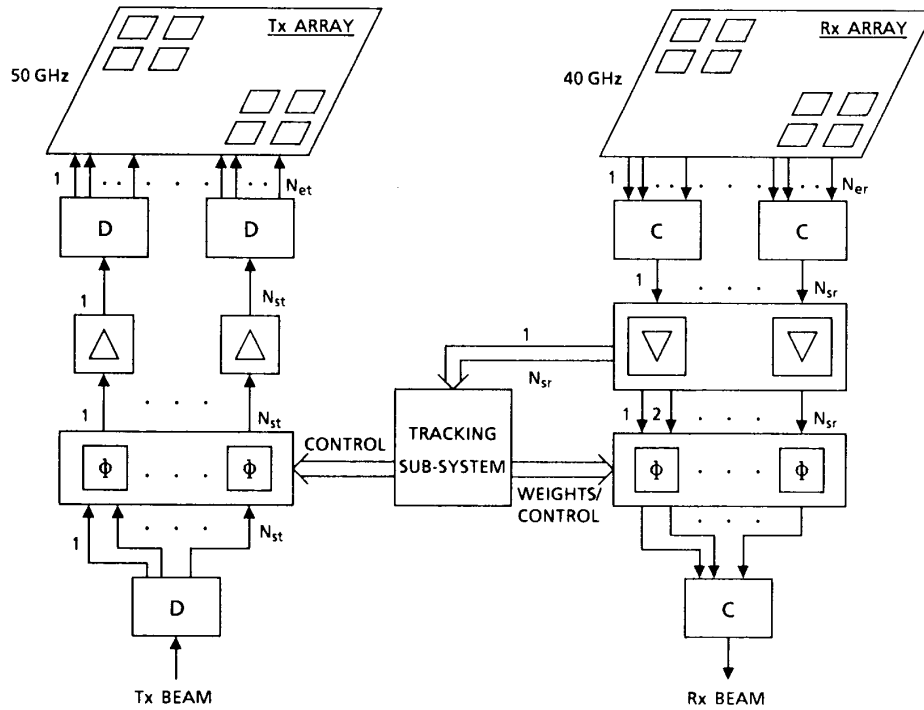


Fig. 10. Block diagram of the mobile terminal array antenna subsystem.

subsystem is proposed in Fig. 10. Square arrays of N_e elements ($\lambda/2$ spacing) are assumed as the basic configuration. The possibility of grouping the N_e elements into equal phase subarrays is also shown. This could be a proper choice for reducing the number of phase shifting and amplifying modules. The tracking subsystem provides control signals to the transmitting and receiving beam-forming networks, which should be realized at radio frequency to lower frequency conversion complexity.

The number of elements per array must be chosen according to the system requirements coming from the link budgets. The terminal G/T and EIRP have been calculated as a function of N_e , assuming an antenna efficiency of 0.55 (scanning loss included), a receiver noise temperature of 350°K, and a transmitter output power of 3W. Table III shows three possible values for N_e which satisfy the link budgets (see Section IV-C) for $d_a = 2$ m, 1 m, and 0.5 m, respectively.

The need for a large number of elements per array calls for an extensive circuit integration. MMIC's have already been successfully exploited in similar antenna configurations [21]. The degree of integration is critical to subsystem cost, assembly, reliability, testability, repairability, and wafer yield. In Fig. 11, the candidate circuits for MMIC integration are shown. The dividing and combining sections are only necessary when the subarray approach is used. The pseudomorphic HEMT on GaAs is a promising device for meeting the 40 GHz noise and 50 GHz power specifications. State-of-the-art performance of hybrid and monolithic amplifiers and phase shifters in the

TABLE III
USER TERMINAL FIGURES OF MERIT AS A FUNCTION OF ANTENNA COMPLEXITY

N_e	G/T (dB/K)		EIRP (dB W)	
	Calculated	Specified	Calculated	Specified
256	0.1	-1.0	30.3	29
1024	5.9	5.0	36.1	35
3600	11.2	11.0	41.4	41

EHF band seem to point out the compatibility of the mobile terminal requirements with medium term noise, power, and phase control performance [22]–[24].

VII. CONCLUSIONS

In this paper, we have indicated a multispot satellite system in the EHF frequency band, having in mind its possible integration with existing terrestrial cellular systems. In spite of the complex behavior of the satellite channel (affected by both propagation effects, environmental effects, and cochannel interference), we introduced a channel model and showed that an achievable target for the system performance could be 99% availability and 10^{-6} bit error probability in rural and suburban areas, with a reduced grade of service within metropolitan areas. To identify more precise channel models and to optimize the system design, results from experiments at EHF with high elevation are needed.

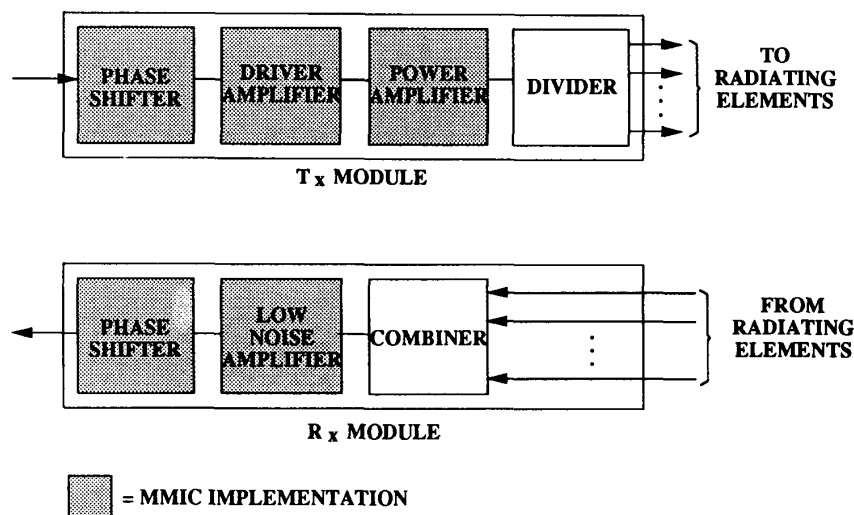


Fig. 11. Possible MMIC integration in the mobile terminal front ends.

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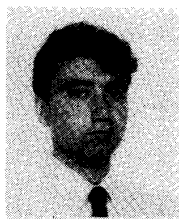
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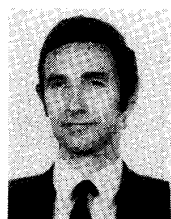
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